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Machine learning-based prediction of laser ablation impulse from plasma spectra

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ABSTRACT

This study explores a machine learning-based approach for predicting the micro-thrust generated by laser ablation of aluminum alloy materials using laser-induced breakdown spectroscopy (LIBS). An impulse-spectroscopy coupled measurement technique is introduced to construct datasets combining plasma spectral information and impulse coupling coefficients under various laser parameters. Using these datasets, support vector machine, and Neural Network models are developed to establish relationships between plasma spectral parameters and the propulsion performance of high-energy pulsed laser ablation. The models demonstrate the ability to predict impulse coupling coefficients based on plasma spectral data, achieving an R^2 value of up to 0.998 for aluminum alloy targets under specific conditions. These findings highlight the potential of LIBS-based methods for measuring laser propulsion performance and provide data-driven insights for understanding the interactions between laser ablation and material properties.

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I. INTRODUCTION

The concept of laser propulsion dates back to the mid-20th century, shortly after the invention of the laser, when scientists began exploring its potential for space propulsion. In the late 1960s, American physicist Arthur Kantrowitz proposed using laser beams as a propulsion mechanism.[1] By the 21st century, advancements in laser technology and materials science facilitated more sophisticated experiments.[2,3] Some projects have since investigated the application of laser propulsion in specific missions, such as microsatellite orbit adjustments and space debris removal.[4–6] NASA and other international space organizations are actively researching laser propulsion, aiming to incorporate this technology into future deep space exploration missions. Compared to conventional propulsion methods, laser propulsion offers several advantages, including high specific impulse, clean emissions, extended operational range, lightweight design, precise control, reusability, and reduced dependence on fuel resupply. These characteristics position laser propulsion as a promising technology for future space exploration and other advanced applications.[7–10]

Laser propulsion technology uses laser beams as a power source to propel spacecraft or other objects. Within this field, laser ablation propulsion has emerged as a critical area of focus. This technology involves directing a high-energy laser beam onto the surface of a solid material, causing localized heating to extremely high temperatures. The intense heat leads to rapid vaporization and ablation of the material, producing a high-speed jet that generates thrust.[11,12]

The three main parameters of laser ablation propulsion performance are impulse I, impulse coupling coefficient C_m, and specific impulse I_{sp}. Impulse I is defined as the integral of the reaction force over time, generated by phenomena such as plasma ejection and material splashing under the high-temperature and high-pressure conditions induced by laser ablation. The impulse coupling coefficient C_m quantifies the efficiency with which the incident laser energy is converted into impulse during the ablation process. It is related to the impulse I and the incident laser energy E. Specific impulse I_{sp} refers to the ability of the propellant to provide impulse per unit mass ablated during the laser ablation propulsion process. It depends on both the ablated mass and the impulse.[13] Among these parameters, impulse I is particularly critical to evaluating the performance of laser ablation propulsion.

Various methods are employed to measure propulsion performance indicators in laser ablation propulsion. Based on differing measurement principles, impulse I measurement can be broadly classified

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into three main categories: the rail method,[14] the impact pendulum method,[15,16] and the sensor method.[17] The rail method is grounded in Newton's second law, the impact pendulum method relies on the dynamic equations of a simple pendulum, and the sensor method utilizes the piezoelectric effect. Among these, the impact pendulum method is the most established technique and is widely used for single-pulse impulse measurements in laser ablation propulsion. However, each of these methods has inherent limitations. The rail method features a complex system and low resistance to interference. The impact pendulum method, despite its maturity, has a complicated structure and slow measurement speed. The sensor method suffers from weak output signals and limited anti-interference capabilities. Additionally, all these approaches are restricted to measuring a single parameter and are unsuitable for on-orbit measurements of thrusters.

In recent years, machine learning has garnered significant attention from researchers across diverse fields, including medicine and materials analysis, leading to notable achievements.[18,19] Its adaptability has also driven a growing trend toward integrating machine learning with thrust measurement techniques. Feng *et al.* applied machine learning to on-orbit measurements of Hall thrusters. By using telemetry parameters as inputs, they predicted thrust with both a radial basis function (RBF) neural network model and the XGBoost (decision tree) model. The RBF neural network achieved a mean square error (MSE) of 0.0914, outperforming the XGBoost model, which had an MSE of 0.1138.[20] Similarly, Wu *et al.* employed an RBF neural network model to predict the thrust of a radio frequency ion thruster. Their input parameters included flow rate, radio frequency power, and acceleration voltage, yielding accurate prediction results.[21]

According to Phipps' research, for nanosecond lasers,[13,22] the primary mechanism of thrust generation is attributed to ion motion at laser energy densities near the optimal impulse coupling level. When the energy density is below this optimal value, thrust generation results partly from the ejection of vaporized material and partly from ion motion. Conversely, at energy densities exceeding this threshold, plasma shielding reduces the amount of laser energy coupled to the target material. This diminishes ion ejection and, consequently, decreases the resulting thrust. These findings highlight the critical role of plasma in impulse generation. Research on the integration of machine learning and plasma modeling is also advancing. Yue ZQ *et al.* demonstrate the efficiency of machine learning in the correction of spectral intensity variations in laser-induced breakdown spectroscopy (LIBS).[23] In our previous studies, we combined laser-induced breakdown spectroscopy (LIBS) with various machine learning algorithms, including Decision Tree (Tree), support vector machine (SVM), Gaussian Process Regression (GPR), and Neural Networks, to simulate time-resolved spectra of copper samples at varying distances from the lens. These simulations demonstrated that machine learning can effectively and accurately predict laser-induced plasma temperatures.[24] Despite these advancements, few studies have explored the intersection of machine learning, laser propulsion, and spectral detection. This gap underscores the need for further research to unlock the potential of machine learning in these cross-disciplinary areas.

This paper focuses on applying machine learning methods to predict the micro-thrust of laser ablation on aluminum alloy materials using laser-induced breakdown spectroscopy (LIBS) technology. A novel impulse-spectrum coupling measurement technique is introduced. Initially, the actual impulse values are measured using a torsion pendulum device, and these are compared with simulation algorithm results. Based on multiple aluminum spectral lines, the normalized coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE) is calculated to evaluate model performance. The objective is to achieve rapid and accurate measurements of the micro-thrust of laser-ablated aluminum alloy materials using LIBS technology. This study not only demonstrates the application of advanced computational techniques for predicting propulsion performance from the spectral information of laser-induced plasmas but also provides a comprehensive comparison of model performance. The goal is to identify efficient algorithmic models that enhance the accuracy and reliability of impulse predictions derived from plasma spectral data.

In summary, this research addresses the limitations of existing laser ablation thrust measurement systems, including poor anti-interference capabilities and slow measurement speeds. By establishing a relationship between plasma spectral information and laser propulsion performance through machine learning, this study offers a faster and more versatile alternative to traditional micro-thrust measurement methods. The findings hold significant value for the rapid measurement and diagnosis of laser propulsion performance parameters.

II. METHOD

A. Experimental methods and apparatus

The experimental setup is illustrated in Fig. 1. A Nd:YAG laser with a wavelength of 1064 nm, a pulse width of 8 ns, and a maximum single-pulse energy of 100 mJ was used, which was operated in the single-trigger mode to ensure the precise control of the plasma evolution time. As shown in Fig. 1, the high-energy laser beam passes through an aperture, a spectroscope, and a mirror before being focused onto the aluminum target surface by a plano–convex lens with a diameter of 12.5 mm and a focal length of 100 mm. The measured focal spot radius was 252.35, 265.61, 276.32, 382.47, 423.68, and 559.72 μm. The aluminum target, mounted at the end of a torsion pendulum, allows simultaneous acquisition of spectral and impulse data. The plasma generated by the laser ablation of the aluminum target is collected using two plano–convex lenses, which focus the light onto a fiber connected to an Avantes 8-channel spectrometer. The spectrometer collects spectral data in the range of 200–1050 nm. The integration time is 1 millisecond. For the experimental timing control, a Stanford DG645 digital delay generator is used to realize the μs-level synchronization between laser triggering and spectral acquisition. A set of background data are collected before the formal test for correction of the formal test data, and the spectral data are processed with noise reduction after the data collection is completed.

At the same time, a torsion pendulum device measures the micro-impulse produced when the laser interacts with the aluminum target. The setup of the impulse measurement system is detailed in the dashed frame of Fig. 1, showing the arrangement of the target material, swing arm, and displacement sensor. When a pulsed laser strikes the target material, an instantaneous impulse is generated, causing the swing arm to oscillate. A displacement sensor, specifically a rangefinder, records the linear displacement of the swing arm. The impulse acting on the torsion pendulum is calculated based on the maximum displacement recorded by the sensor, in combination with the physical parameters of the torsion pendulum device, such as moment of inertia, moment arm, measuring arm, and damping ratio. The impulse calculation formula is derived according to the principles described in Ref. 25, as follows:

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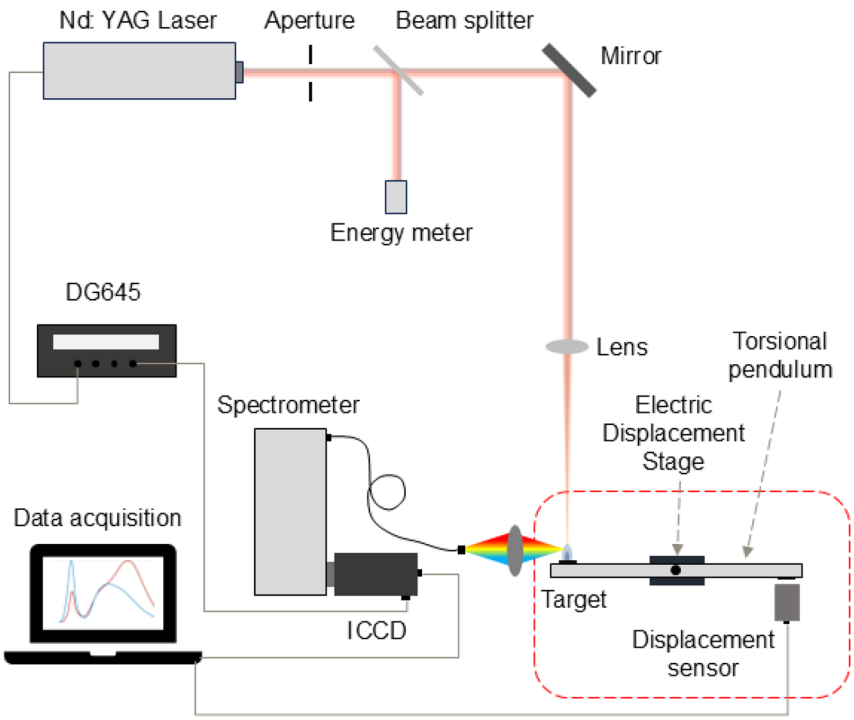

FIG. 1. Experimental setup diagram of the impulse-spectral coupling measurement system.

$$I = \frac{J\omega_n}{L_f L_m} \exp\left(\frac{\zeta}{\sqrt{1-\zeta^2}} \arctan \frac{\sqrt{1-\zeta^2}}{\zeta}\right), \tag{1}$$

where I, J, and ω_n denote impulse, moment of inertia, and natural angular frequency, respectively; L_f, L_m, and ζ denote other relevant parameters. The test system is capable of measuring impulses ranging from 10^{-7} to 10^{-5} N·s.[26] To ensure data accuracy, each experimental configuration was repeated five times, with all laser energy levels measured multiple times. The laser energy uncertainty was maintained below 2%. An electrodynamic displacement table was installed beneath the torsion pendulum to ensure that each laser ablation event illuminated a fresh surface on the target material.

During the experimental process, laser energy was precisely adjusted using a half-wave plate and a Glan prism mounted on a high-precision rotary stage. A Coherent laser energy meter was employed to measure the laser energy. For each rotation angle of the wave plate, laser energy measurements were conducted five times, and the average value was recorded. By varying the rotation angle of the wave plate, 55 distinct energy levels were achieved, ranging from 26 to 88 mJ. Spot sizes were controlled by adjusting the distance between the lens and the target material, resulting in six different spot sizes. The corresponding power density ranged from 4.68×10^8 to 5.52×10^9 W/cm^2. To enhance the reliability of the data, each experimental configuration was repeated five times, yielding a total of 1650 experiments and 330 sets of valid data.

B. Selection of models

Two commonly used regression models were selected to simulate and predict the impulse generated by laser ablation of aluminum targets:

(1) Support vector machine (SVM):[27] SVM is a supervised learning model with an associated algorithm, widely used for nonlinear regression problems. As one of the most common kernel-based learning methods, it offers advantages in robustness and simplicity, particularly when addressing challenges related to small sample sizes, nonlinearity, and high-dimensional pattern recognition.
(2) Neural Network:[28] Neural Networks are a class of algorithms inspired by the structure and function of the human brain, designed to recognize patterns and interpret sensory data. They excel in processing large datasets and can automatically learn the representations needed for feature detection, making them highly effective for clustering or labeling raw inputs.

For this study, regression programs were developed in Python using algorithms from the sk-learn library. Model parameters were optimized through grid search to identify their optimal values.

III. RESULTS AND DISCUSSION

A. Plasma spectrum and impulse

Figure 2 presents the spectral data obtained after laser irradiation of an aluminum target. In addition to the spectral lines of aluminum, lines corresponding to oxygen and nitrogen are also observed. These features result from the high-temperature and ionization effects caused by the focused high-energy laser beam. Specifically, air breakdown occurs when the laser energy is sufficiently high, ionizing air molecules—primarily nitrogen and oxygen—to form plasma, which emits characteristic spectral lines. Simultaneously, the laser interaction with the aluminum target surface leads to vaporization, melting, and ablation. At elevated temperatures, aluminum reacts with oxygen in the air to form aluminum oxide, releasing oxygen and volatile products that

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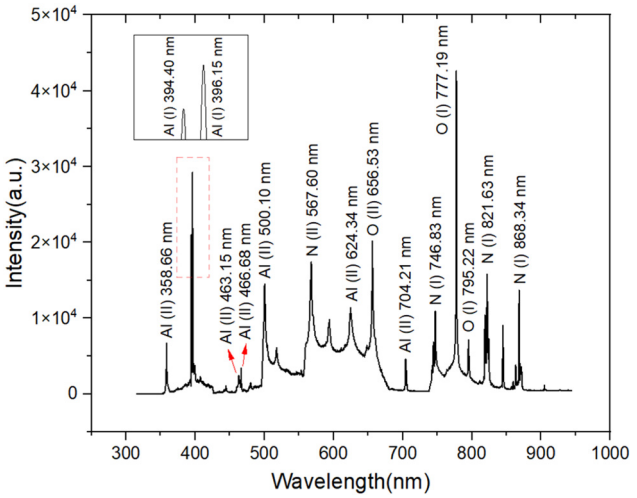

FIG. 2. Typical spectrum of laser-induced plasma from an aluminum target (pulse energy 88 mJ).

contribute to the spectral features. Furthermore, the expanding plasma contains particles ejected from the aluminum sample as well as ionized gas molecules from the surrounding air. As the plasma cools, the spectrum reflects contributions from both the aluminum sample and the air components, providing valuable information for subsequent analysis.[29]

Figure 3(a) shows the emission spectra of aluminum plasma at different laser energies, illustrating significant changes in spectral intensity as the laser energy varies. It is evident that the spectral intensity increases significantly with higher laser energy. On the other hand, Fig. 3(b) depicts the variation in impulse generated by the aluminum target under laser irradiation, showing a similar trend of increasing impulse with increasing laser energy. Figure 3(c) shows the relationship between the intensity values of different aluminum spectral lines and impulse, and it can easily found that impulse and spectral line intensity are positively correlated. From the whole Fig. 3, it seems that the intensity of the Al(I) at the 394.4 nm line increases with laser energy until it reaches a plateau. This is typical behavior for lines that suffer from self-absorption. The spectral self-absorption phenomenon produced by the laser-induced breakdown of the aluminum target is essentially a photon resonance reabsorption effect due to the nonuniformity inside the plasma. When a high-energy laser pulse is applied to the aluminum surface, the high-temperature and high-density plasma formed instantaneously forms a significant temperature and density gradient distribution during the expansion and cooling processes, i.e., the core region of the plasma maintains a high-temperature and high-excited state, while a large number of ground-state aluminum atoms accumulate in the peripheral low-temperature region (<8000 K). When the excited state aluminum atoms in the core region jump to the ground state to emit characteristic photons, these photons will be resonantly absorbed by the same ground-state atoms in the peripheral low-temperature region in the outward propagation process, resulting in the intensity of the wavelength at the center of the emission spectra decaying, or even the center of the spectral line being concave (self-reversal phenomenon). However, since this study mainly learns spectral features, the aforementioned self-absorption phenomenon does not affect the impulse prediction performed by machine learning.

During the laser-induced plasma process, the interaction between the laser and the material leads to ionization and plasma formation, which produces characteristic spectra. Simultaneously, the explosive expansion of the plasma, combined with the principle of momentum conservation, results in a recoil force on the target material, generating an impulse. The relationship between spectral intensity and impulse magnitude at different laser energies is presented in Fig. 3, demonstrating a strong correlation between the two. Notably, the trend in impulse variation aligns closely with the trend in spectral intensity variation. This phenomenon can be explained by the following mechanisms: when a high-energy laser pulse interacts with the surface of a material or a gas, the strong electric field induces ionization, forming plasma through mechanisms, such as multiphoton ionization, collisional ionization, and thermal ionization.[30] Once plasma forms, its particles—such as electrons, ions, and neutral atoms—are in highly excited states. As these particles transition to lower energy levels, they emit photons at specific wavelengths, producing an emission spectrum.[31] The intensity, position, and width of these spectral lines provide insights into the plasma's temperature, density, and composition. For example, high-temperature plasmas exhibit broader and more intense spectral lines.

In addition to electromagnetic radiation, laser-induced plasma also involves momentum transfer and mechanical effects. The characteristic explosive expansion, conservation of momentum, and thermally driven shock waves all contribute to impulse generation. During explosive expansion, the laser pulse causes rapid heating and vaporization of a localized region of the target, creating a high-pressure plasma. This high-pressure zone expands outward, generating a pressure difference with the surrounding environment, leading to explosive plasma expansion. The resulting reaction force on the target material generates an impulse in the opposite direction, consistent with Newton's Third Law. Regarding momentum conservation, a portion of the laser photon's momentum is transferred to plasma particles. According to the law of conservation of momentum, the target gains momentum in the opposite direction to the plasma, producing an impulse. Furthermore, as shown in Fig. 4, the rapid heating and expansion of the plasma induce a shock wave in the surrounding medium. This shock wave carries significant kinetic energy and exerts a mechanical impact on the surrounding environment, further contributing to impulse generation. Overall, the plasma spectrum and impulse are intrinsically linked, with changes in one reflecting corresponding changes in the other. This relationship underscores the importance of plasma spectral analysis for understanding the dynamics of laser-induced impulses.

B. Data preprocessing

In this study, the training and validation sets were constructed using a stratified random splitting strategy in strict compliance with machine learning best practice guidelines. Specifically, the dataset is divided into 80% training set (for model parameter optimization and feature learning) and 20% independent test set (for final performance evaluation), the splitting process is implemented by the train test split function of the Scikit-learn library, and the random state is set to fix the random seed to ensure the reproducibility of the experiments.

Ideally, time-gated detection with a short gate width is preferred to accurately capture the early plasma dynamics. However, due to the

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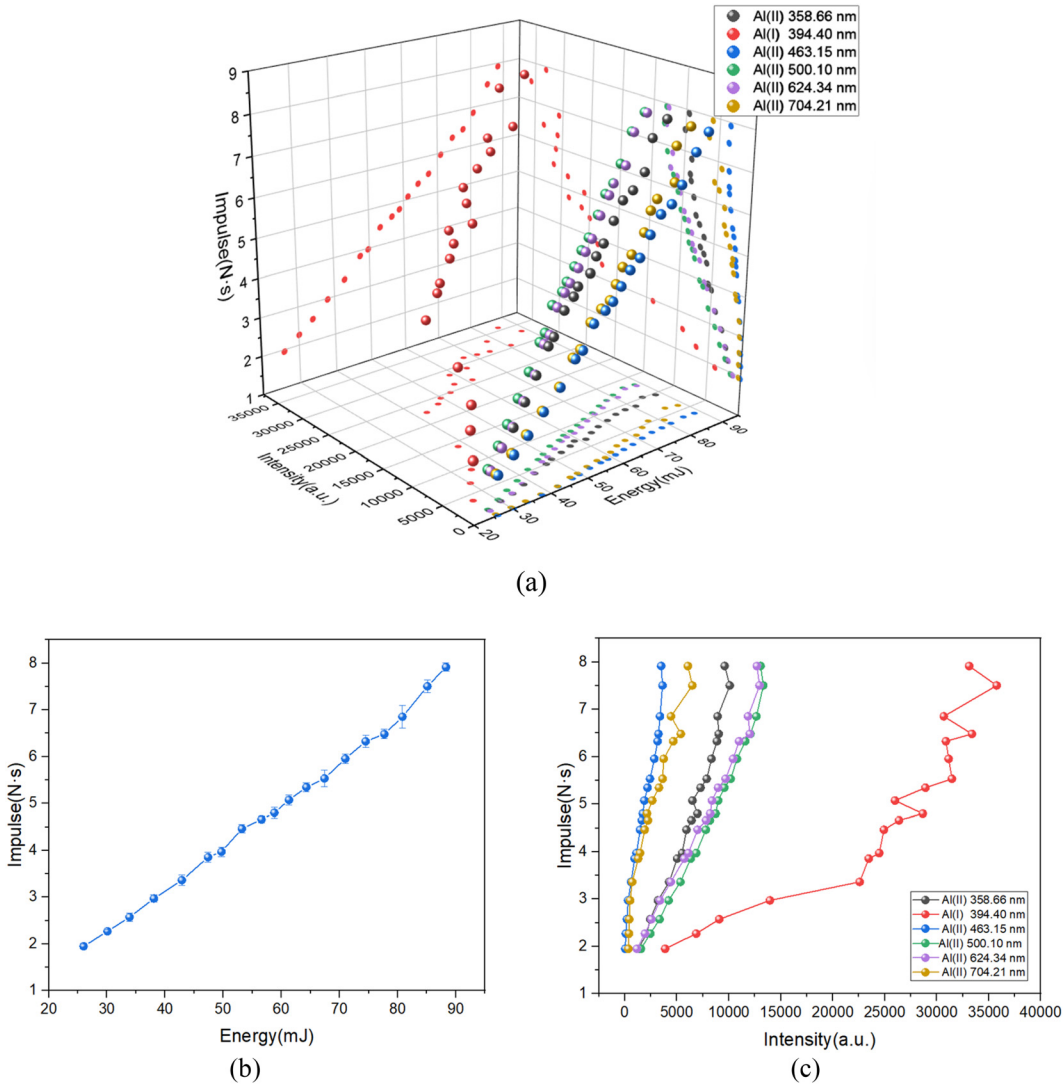

FIG. 3. When the focused laser spot radius is 265.61 μm, (a) the relationship between energy, intensity of emission spectra, and impulse. (b) Impulse generated by laser irradiation of an aluminum target as a function of laser energy. (c) Impulse generated by laser irradiation of an aluminum target as a function of spectral intensity at different aluminum spectral lines.

limitations of the spectrometer used in this study, time-gated detection is currently not feasible, and we had to rely on integration-based spectral acquisition. Although longer integration times are generally not conducive to maintaining local thermodynamic equilibrium (LTE), they can still be used to characterize the overall emission trends of the plasma. As shown in Fig. 3, there exists a positive correlation between the plasma impulse and spectral intensity. Hence, we utilize spectral intensity measurements to qualitatively reflect the changes in plasma impulse. In practice, we select emission lines that are highly sensitive to variations in laser energy and exhibit a strong signal-to-noise ratio (as depicted in Figs. 2 and 3).

To prepare the data for machine learning, spectral line peaks were identified programmatically and matched with characteristic spectral wavelengths in the NIST database. Eight spectral lines with the highest intensities were selected as the primary feature values for the training set. These lines include Al II 358.66, Al I 394.40, Al II 463.15, Al II 500.10, Al II 624.34, and Al II 704.21 nm. Additional feature values, such as laser energy, were also incorporated. The dataset was normalized to ensure that the magnitudes of various attributes in the input vector were comparable. Large discrepancies in attribute magnitudes can adversely affect the prediction results, leading to distortions.[32] By scaling the data to a consistent range, normalization enhances the algorithm's convergence speed and improves the overall model performance.[33] We employed the min-max scaling method to normalize the data, with the formula as follows:

$$X_{new} = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}}. \tag{2}$$

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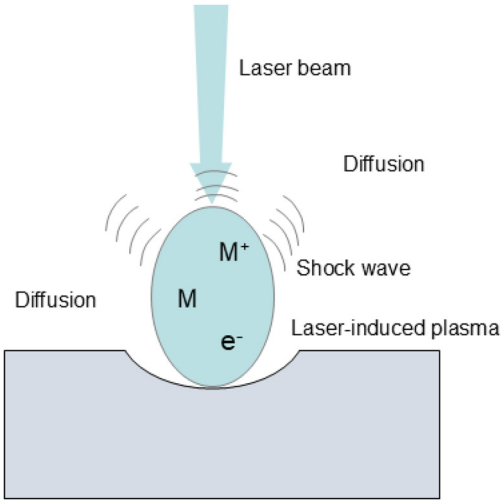

FIG. 4. Plasma expansion and shock wave formation during the laser-induced plasma process.

This preprocessing step (1) preserves original data distribution patterns while (2) constraining all features to [0,1] range to ensure equal weight in model training. Additionally, we developed a program to perform denoising, wavelength matching, and peak extraction on the spectral curves, converting the spectral intensities into a two-dimensional array to address the matching between energy values and spectral intensities.

C. Model training and evaluation

The training and evaluation of the model was conducted through two sequential steps. First, the dataset was partitioned into a test set and a validation set. During model validation, we further subdivided the test set into subsets for cross-validation. After model training was completed, comprehensive testing was performed on the independent test set. Crucially, this test set remained entirely separate from the data used in cross-validation, thus providing a more accurate assessment of the model's generalization capability.

1. Model optimization strategy

We employed randomized search for both model optimization processes. For the SVM, three kernel functions (linear, rbf, and poly) were configured, with hyperparameters sampled from log-uniform distributions. A total of 50 parameter combinations were evaluated using fivefold cross-validation. For the neural network, architectures with varying layers and neuron counts (single-layer: 50 or 100 units; two-layer: 50×50) were tested. L2 regularization was applied to prevent overfitting. A two-phase search strategy was implemented: an initial coarse-grained search with 30 parameter combinations and threefold cross-validation, followed by a refined search of 15 combinations using fivefold cross-validation based on preliminary results.

2. Model evaluation

This study begins by examining the influence of various input parameters on prediction performance, identifying several key indicators such as laser energy and specific aluminum spectral lines, as summarized in Table I. Subsequently, these well-performing indicators were combined to further enhance prediction accuracy, with the results shown in Table II.

TABLE I. Comparison of R^2, RMSE, and MAE values for the training set of the two regression models when a single input parameter is used.

	SVM			Neural network		
Input parameter	R-squared	RMSE	MAE	R-squared	RMSE	MAE
Energy	0.937	0.429	0.331	0.944	0.407	0.348
Al II 358.66 nm	0.764	0.834	0.661	0.750	0.858	0.706
Al I 394.40 nm	0.744	0.868	0.612	0.743	0.869	0.686
Al II 463.15 nm	0.965	0.322	0.211	0.956	0.361	0.277
Al II 500.10 nm	0.734	0.885	0.720	0.726	0.898	0.748
Al II 624.34 nm	0.779	0.806	0.627	0.789	0.787	0.638
Al II 704.21 nm	0.957	0.356	0.271	0.943	0.410	0.335

To assess the accuracy improvements achieved by the prediction models, several commonly used statistical metrics in regression analysis were calculated. These metrics provide a comprehensive evaluation of model performance. The calculation formulas for these indicators are as follows:[34]

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}, \tag{3}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}, \tag{4}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{5}$$

Table I presents the calculation results of the two models when using a single variable as the input parameter. It is important to note that R^2 represents the coefficient of determination, where values closer

TABLE II. Comparison of R2, RMSE, and MAE values for the training set of the two regression models when multiple parameters are used.

	SVM			Neural network		
Input parameter	R-squared	RMSE	MAE	R-squared	RMSE	MAE
Two lines	0.994	0.128	0.081	0.990	0.171	0.114
Three lines	0.996	0.103	0.070	0.992	0.153	0.099
Four lines	0.993	0.143	0.104	0.990	0.173	0.113
Five lines	0.997	0.099	0.069	0.992	0.154	0.104
Six lines	0.997	0.087	0.065	0.987	0.196	0.131
Energy + 1line	0.998	0.079	0.063	0.986	0.202	0.168
Energy + 2lines	0.998	0.064	0.052	0.993	0.139	0.100
Energy + 3lines	0.999	0.065	0.055	0.995	0.126	0.091
Energy + 4lines	0.998	0.072	0.059	0.995	0.115	0.085
Energy + 5lines	0.999	0.064	0.054	0.994	0.136	0.095
Energy + 6lines	0.998	0.068	0.057	0.993	0.148	0.109

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to 1 indicate better performance. RMSE, on the other hand, refers to the root mean square error, which quantifies the discrepancy between the actual and predicted values; hence, smaller RMSE values are preferable. From Table I, it is evident that when using Al II 463.15 as the input parameter, both the SVM and Neural Network models achieve relatively optimal results, with the key indicator R^2 reaching as high as 0.965 and 0.956. This highlights the significance of Al II 463.15 as an input variable in achieving accurate predictions.

Considering the combination of input parameters may achieve better prediction results, Table II presents a comparison of R^2, RMSE, and MAE values for different regression models when the input parameters are combined variables. Upon careful examination of

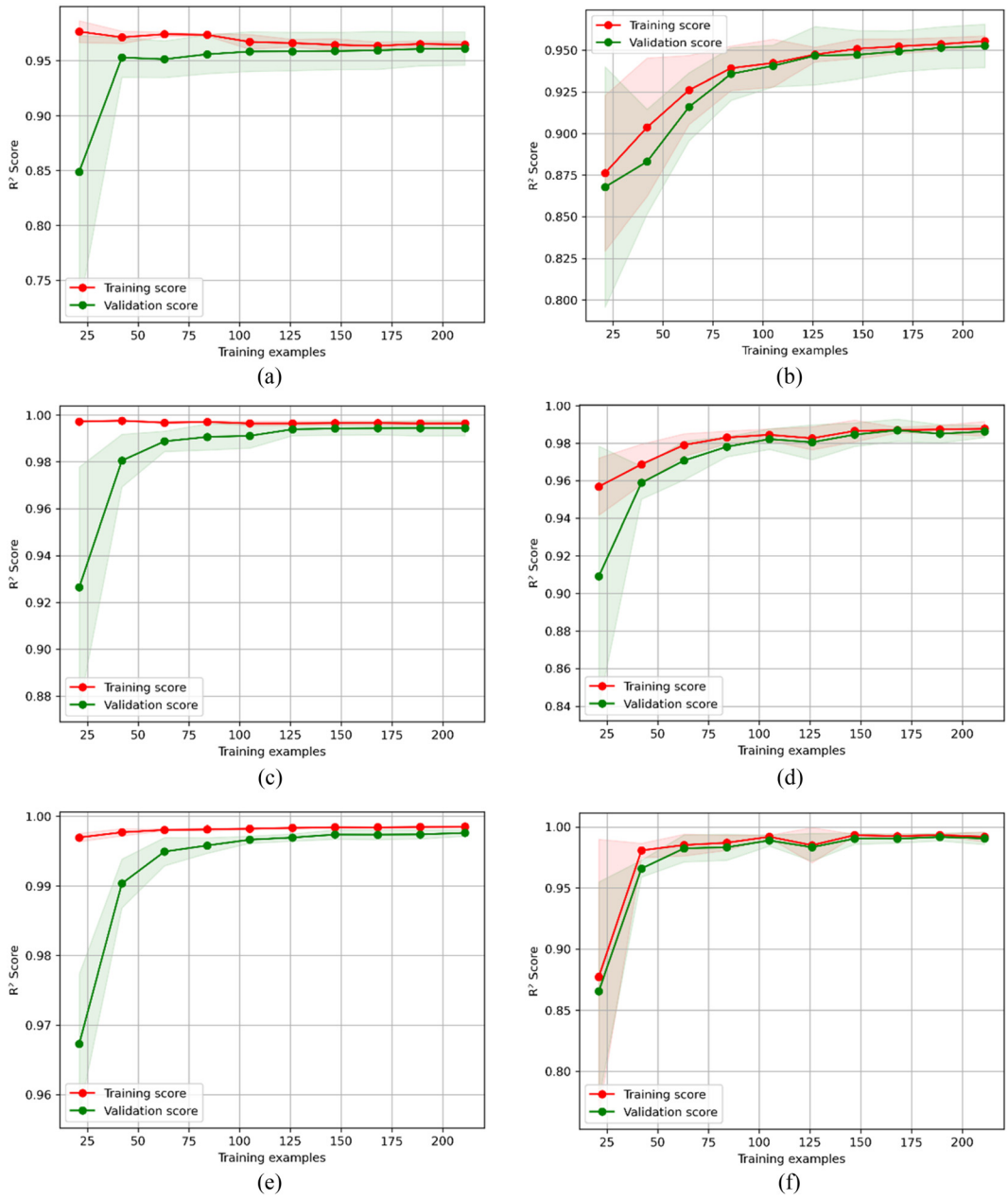

FIG. 5. Learning curves of the SVM model and the neural network model during the training process. The top-performing models were selected for three input parameter configurations: single spectral line, multiple spectral lines, and energy combined with multiple spectral lines. (a) SVM—Al II 463.15 nm, (b) Neural Network—Al II 463.15 nm, (c) SVM—358.66, 463.15, and 500.10 nm, (d) Neural Network—Al II 358.66, 463.15, 500.10, 624.34, and 704.21 nm, (e) SVM—Energy + Al II 358.66, 463.15, and 500.10 nm, (f) Neural Network—Energy + Al II 358.66, 394.40, 463.15, 500.10, and 624.34 nm.

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Table II, it can be observed that using multiple spectral lines for prediction yields more accurate results as the number of spectral lines increases. When combined with energy for prediction, the accuracy of R^2 can even reach 0.999. However, it should be noted that the optimal prediction scheme corresponding to different models is slightly different. The SVM model is in the case of energy combined with 3 spectral lines as input parameter, while the NEURAL NETWORK model is in the case of energy combined with 5 spectral lines as input parameter, which is mainly caused by differences in different algorithms. When comparing MAEs further when R^2 is used as a measure, it can be seen that the NEURAL NETWORK model is slightly inferior to the SVM model. In practical prediction, it is necessary to select input parameters based on their specific characteristics to achieve more accurate prediction results.

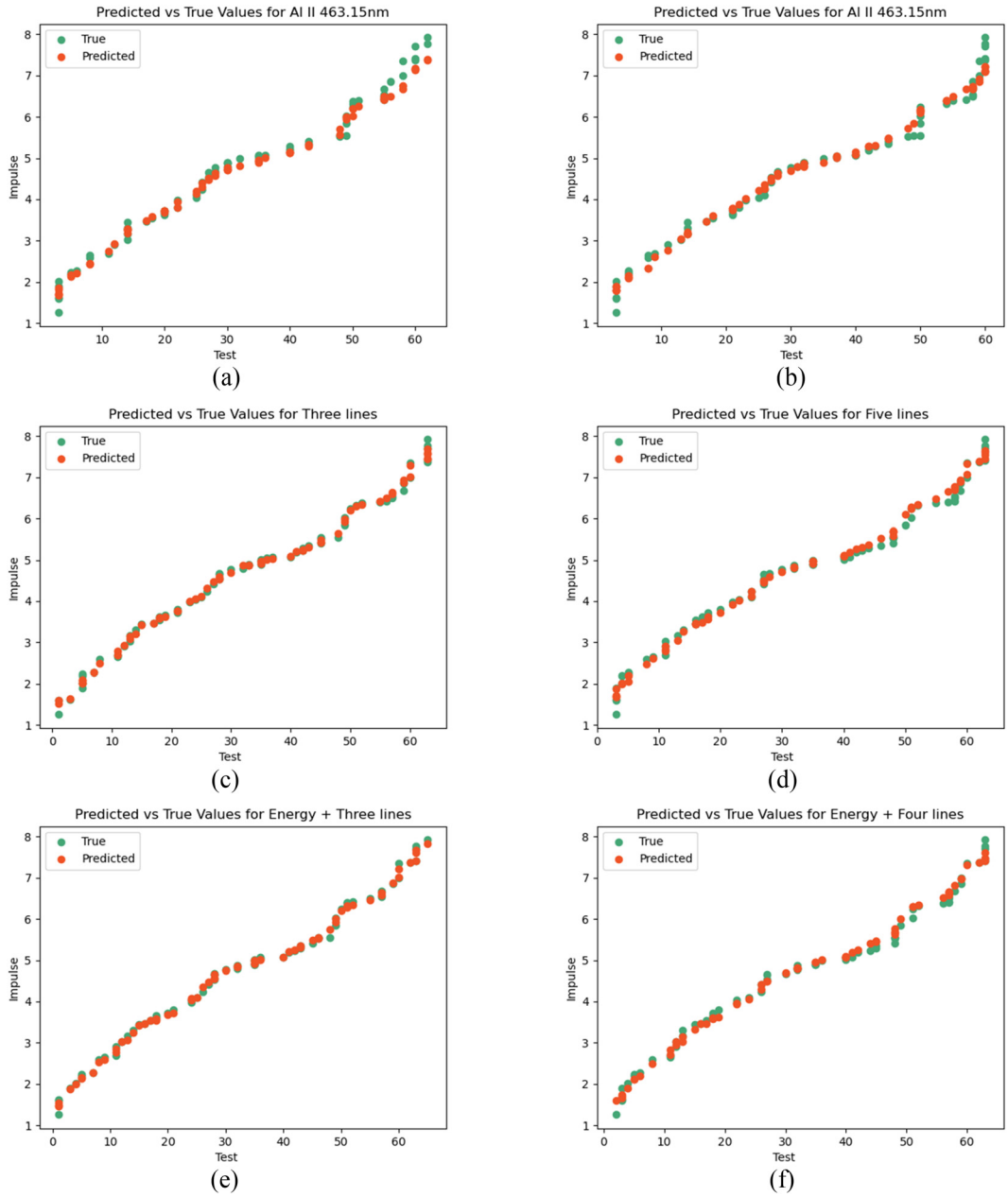

FIG. 6. Comparison of impulse values predicted by the SVM model and Neural Network model vs experimentally measured values. (a) SVM—Al II 463.15 nm, (b) Neural Network—Al II 463.15 nm, (c) SVM—358.66, 463.15, and 500.10 nm, (d) Neural Network—Al II 358.66, 463.15, 500.10, 624.34, and 704.21 nm, (e) SVM—Energy + Al II 358.66, 463.15, and 500.10 nm, (f) Neural Network—Energy + Al II 358.66, 394.40, 463.15, 500.10, and 624.34 nm.

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Figure 5 evaluates the models using R^2 as the scoring criterion, with the green line representing the cross-validation score. The figure shows that for both models, the learning curves converge rapidly as the sample size increases, indicating that the SVM model and the Neural Network model exhibit strong adaptability to the measured dataset. In summary, both the Neural Network model and the SVM model demonstrate excellent performance in predicting laser ablation impulse based on plasma spectral information.

D. Model testing

In Sec. III C established a reliable model and identified optimal combinations of input parameters that, when used with the model, yield improved prediction results. To validate the models, Fig. 6 presents the comparison of impulse values predicted by the SVM model and Neural Network model vs experimentally measured values. The top-performing models were selected for three input parameter configurations: single spectral line, multiple spectral lines, and energy combined with multiple spectral lines. As shown in Fig. 6, the predicted values closely align with the experimentally calculated values in most schemes, demonstrating excellent prediction accuracy. This agreement highlights the effectiveness of both the Neural Network and SVM models in predicting impulse values based on the chosen input parameters.

To clarify this information more clearly, Fig. 7 illustrates the impact of different input parameters on the R-squared value, with error bars indicating the 95% confidence interval range. This visualization reflects both the accuracy and stability of the model predictions. It can be observed that as the input parameters are adjusted, the R^2 value increases until reaching a stable level, while the confidence interval progressively narrows. This pattern indicates that the model predictions increasingly align with actual values while demonstrating enhanced predictive stability, signifying continuous performance improvement. Additionally, comparative analysis reveals that the SVM model exhibits slightly superior performance compared to the neural network model.

IV. CONCLUSION AND DISCUSSION

This study focuses on predicting the impulse generated by laser ablation of aluminum targets using machine learning algorithms and plasma spectral information. First, the intrinsic relationship between spectral line intensities and impulse magnitudes was analyzed based on experimental data. During the laser-induced plasma process, the interaction between the laser and the target material leads to ionization and plasma formation, producing characteristic spectra. Simultaneously, the explosive expansion and momentum conservation effects result in a recoil force on the target, generating impulse. These two phenomena are inherently linked. Building on this theoretical foundation, the prediction performance of different machine learning models, including the SVM model and the Neural Network model, was compared. The results show that both models achieve excellent prediction accuracy under various input parameter schemes. When evaluated using R^2, RMSE, and MAE, the SVM model demonstrated superior performance, achieving an R^2 of 0.999, an RMSE of 0.064, and an MAE of 0.054. Additionally, using R^2 as a scoring criterion, the learning curves for both models showed rapid convergence as the sample size increased, indicating strong adaptability and applicability for impulse prediction when optimal input parameter schemes are employed. This study develops a rapid and versatile method for predicting the impulse generated by laser ablation of aluminum alloys using spectral information and machine learning. Compared to traditional micro-thrust measurement methods, this approach is faster, more flexible, and potentially fault-tolerant.

The simultaneous acquisition of plasma spectral lines has become increasingly feasible with the maturation of multi-spectral imaging technology. Leveraging this advancement, our research enables non-contact, rapid impulse prediction. However, our current manuscript has focused exclusively on aluminum target measurements under

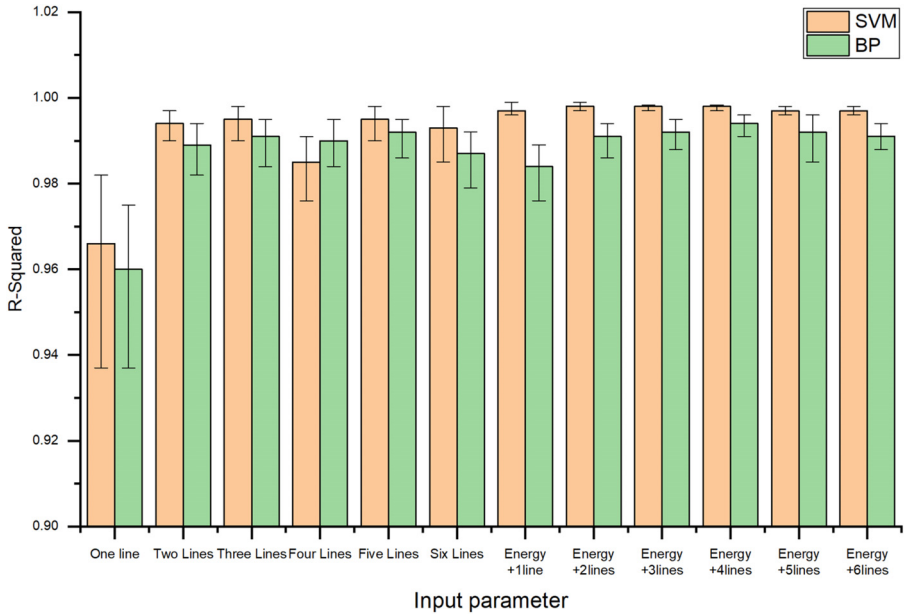

FIG. 7. The impact of input parameters on the R-squared of the test set. The error bars in the figure represent the 95% confidence interval range of the test set.

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controlled laboratory conditions. However, this technique presents some significant limitations: it requires extensive ground-based testing to acquire sufficient data for model development. Furthermore, this study did not thoroughly investigate the model's generalization capability. Under atmospheric conditions, the expansion of the plasma plume, the behavior of shock waves, and momentum transfer are all strongly influenced by interactions with the surrounding air. In contrast, in a vacuum, the plasma expands freely, and there are significant differences in the magnitude of impulse generated under these two conditions. While the method presented in this article, which utilizes plasma spectroscopy and machine learning techniques to predict impulse, is applicable to space environments, the data presented in this article are not suitable for direct application in space.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Heyan Gao: Data curation (equal); Investigation (equal); Software (equal); Supervision (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Ying Wang:** Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jifei Ye:** Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal). **Anmin Chen:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Resources (equal); Software (equal); Validation (equal). **Ming Wen:** Resources (equal); Visualization (equal); Writing – review & editing (equal). **Diankai Wang:** Project administration (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Bangdeng Du:** Investigation (equal); Writing – review & editing (equal). **Sai Li:** Resources (equal); Software (equal); Supervision (equal). **Qianqian Cui:** Formal analysis (equal); Writing – original draft (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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